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1. An SFR is safer than an LWR because:
 - a) It has a higher margin to boiling
 - b) The thermal conductivity of ~~fuel~~ metal fuel is greater, thus cooler fuel temperatures
 - c) It can be operated at atmospheric pressure
 - d) The sodium coolant has more thermal inertia, can protect during an over-power event

- negative reactivity feedback

2. There are many reasons to alloy uranium for metal fuels:
 - a) High temperature cubic γ -phase preferable, alloying can lower temperature for phase transition and stabilize it
 - b) Can raise melting temperature of fuel
 - c) Can introduce minor actinides to be recycled from spent fuel
 - d) Can increase power and fission rate by adding Pu
 - e) Can increase the thermal conductivity of fuel
 - f) Can decrease diffusion rates by adding Zr, leads to less recombination redistribution
 - g) Can mitigate irradiation swelling

3. Metal fuel in an SFR needs a low smear density and large plenum to mitigate FCMI. The fuel burns up and produces fission gas bubbles which increase the volume of the fuel, causing it to come in contact with the cladding. This contact can strain and fail the cladding. However, once the fuel slug swells to about +30% volume, the gas bubbles are released through the development of an open network of porosity. Thus, if we give sufficient room for the fuel to swell before free release of the fission gas, which would be about 75% S/D, and with a large enough plenum to contain gas, then the pressure on the cladding can be reduced to effectively eliminate FCMI.

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4. Many phases occur in metallic fuels, and their constituents continuously redistribute depending on temperature gradients, chemical potential gradients, and solubility. For instance, irradiation can create porosity which degrades thermal conductivity, burnup leading to larger temperature gradients. These gradients can affect the solubility of elements in certain phases, and can drive certain elements up or down in self diffusion. Some phases are only possible at certain temperatures, and the solubility of some elements may be reduced in these phases, causing it to reconstitute as precipitates or diffuse elsewhere.

One example occurs in U-Zr fuel. After some time and burnup, Zr will diffuse up the temperature of the gradient to the hotter center. This leaves an intermediate zone with less Zr, and U diffuses to enrich it. The coolest periphery is basically as-fabricated.

These phases and redistribution affect mechanical properties, thermal conductivity, melting temperatures, size and morphology of fission gas bubbles, and FCCI.

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5. Thermal conductivity degrades with burnup in metallic fuel as fission gas bubbles are produced. However, some noble metal and rare earth precipitates can form that locally increase thermal conductivity. However, once the open porosity network develops and gas is released, bend sodium can infiltrate and subsequently raise thermal conductivity.

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6. FCCI is fuel cladding chemical interaction, and is usually localized. High temperatures and fission product inventories can promote chemical reactions between cladding and fuel, especially Ni, Cr, Fe in cladding with La, Nd, Ce and other rare earth fission products. Cracks in fuel near cladding create large local interaction zones for interdiffusion, like U_2Fe . FCCI can lower melting temperatures and reduce mechanical properties for both the fuel and cladding. Lower melting temperatures mean lower coolant operating temperatures, so the fuel cannot operate at high power.

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7. In MOX fuel, high temperatures and present porosity in fabrication lead to significant recombination early in life. Voids coalesce into a central void at the center of pellets. Small voids are very mobile, but larger voids get stretched into "lenticular" voids. They move toward the hotter center by an evaporation-condensation mechanism. Since the temperature gradient in voids is greater than the pellet, the hotter material at the hotter end evaporates and condenses on the cooler side as a single crystal. This creates a region

of columnar grains around the central void. Past this region where temperatures are lower, the grains become equiaxed and grow this way. At the edge of the pellet, the microstructure is basically as-sintered.

8. Since PuO_2 is not as volatile as UO_2 , it goes along for the ride in the void transport process and is redistributed all along the edge of the central void in the pellet.

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There are also plutonium aggregates in the pellet due to incomplete mixing of mix. As fissions occur, oxygen is freed from its bonds with U/Pu and travels through solid and vapor diffusion down the temperature gradient to the periphery. At the periphery, the O/M ratio approaches 2 in the hypostoichiometric pellet.

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9. JOG is joint oxide gain. Volatile fission products like Cs and I diffuse to the periphery and react with Mo and O to form an oxide. This oxide forms on the cooler outer surface of the pellet (since the oxide is volatile) and forms a mechanical buffer between the fuel and cladding. However, this JOG can also exacerbate cladding corrosion on the inner cladding surface.

10. Sodium corrosion occurs two ways:

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Type I) Preferential dissolution of material constituents (i.e. Ni, Fe, Cr). This depends on the temperature of coolant, saturation concentrations, immersion time, coolant velocity, and impurities present. In a thermal transient, material can be steadily dissolved and redeposited from hotter to cooler areas. Ni is very soluble in Na.

Type II) Impurities in the coolant can corrode materials, such as C, N, and O. Oxygen can increase the solubility of certain elements like Fe. Carbon can transfer into and out of material: carburization and decarburization. Higher Cr alloys tend to carburize since they can form chromium carbides. Lower Cr alloys tend to decarburize.